AYUSH SONI

Senior Data Scientist | Applied ML, Causal Inference, GenAI & Evaluation

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SUMMARY

Senior Data Scientist with 6+ years building applied ML, causal measurement, forecasting, NLP, and grounded GenAI systems across product, finance, and operations. Leads work from problem framing and point-in-time data through evaluation, deployment controls, monitoring, and stakeholder adoption. Recent work includes Snowflake AI agents, governed Text2SQL, 26-week demand planning, contract intelligence, and public model-evaluation infrastructure.

CORE TECHNICAL SKILLS

Applied ML & Causal Modeling: Causal Inference, Quasi-Experimental Design, A/B Testing, SARIMAX, XGBoost, Regression, Classification, Calibration, Ranking

Evaluation & Decision Science: Time-Aware Validation, Success Metrics & Guardrails, Statistical Inference, Threshold Economics, Error Analysis, Business Significance

Programming & Data: Python, SQL, R, Snowflake, Databricks, PySpark, ETL/ELT Pipelines, Git/GitHub

GenAI & NLP: RAG, Reranking, Text2SQL, Tool Calling, Snowflake Cortex AI, Snowflake Intelligence, FinBERT, Hugging Face, LLM Evaluation

MLOps & Cloud: MLflow, FastAPI, Docker, GitHub Actions, CI/CD, Monitoring, Airflow, Jenkins, AWS, GCP, Azure

Visualization & Communication: Tableau, Power BI, Executive Dashboards, KPI Design, Technical Narratives

Leadership: Cross-Functional Delivery, People Management, Mentoring, Code and Analytical Review

PROFESSIONAL EXPERIENCE

DISH / EchoStar (Boost Mobile) — Denver, CO

Senior Data Scientist — Sep 2024 – Present

Manages and mentors team members while leading the AI/ML lifecycle for Retail Wireless Product Data & Analytics. Partners with product, marketing, finance, supply chain, and operations leaders to translate pricing, promotion, inventory, and customer questions into measurable decisions.

- Designed and executed a 4-week quasi-experimental analysis of iPhone Flash Pricing, controlling for weekday mix, SSI, trend, seasonality, and other confounders; quantified +27% GA lift for one device launch and +1,083% for another and identified a structural web-conversion constraint.
- Built a 26-week demand-planning system using SARIMAX and XGBoost, rolling-origin backtests, and hierarchical reconciliation; reached about 87% channel-level and 75% activation-level accuracy and exceeded the 75% N-4 target.
- Built a Snowflake Cortex AI promotion-effectiveness agent that accepts natural-language inputs, invokes audited Python analysis, removes weekday, seasonality, trend, holiday, SSI, and launch effects, and returns decision-ready lift evidence.
- Developed Snowflake Intelligence and Text2SQL workflows over roughly 50 curated business tables from about 200 sources, with governed metrics, relationships, synonyms, verified queries, follow-up analysis, charts, logs, and user feedback.
- Built an OEM contract intelligence agent with version-aware retrieval, evidence-backed answers, citation checks, access controls, and refusal when supporting contract language is missing.
- Architected a four-phase Snowflake deployment framework covering dataset onboarding, staging validation, IT-reviewed YAML promotion, monitoring, and feedback loops for reliable analytical products.
- Built executive Tableau and Snowflake analytics for inventory, activations, pricing, and subscriber outcomes, improving decision speed by 30% and reducing manual reporting effort by 40%.
- Mentors team members on analytical design, implementation quality, and stakeholder communication, and reviews work to establish consistent standards for decision-ready analysis.

The Partnership Center Ltd. — Cincinnati, OH

Data Scientist — Jun 2024 – Sep 2024

- Deployed a Python regression workflow for resource-utilization forecasting, reducing forecast error by approximately 20% and improving schedule-tracking efficiency by approximately 40%.
- Migrated enterprise data to Snowflake and automated Alteryx and Snowflake ETL workflows for repeatable portfolio-performance monitoring.
- Built Power BI dashboards with advanced DAX calculations for investment-portfolio trends and resource-planning decisions.
- Partnered with business stakeholders to operationalize model outputs into recurring resource-planning workflows.

UBS — Chicago, IL

Data Scientist — Quantitative Research & NLP/AI — Jan 2023 – May 2024

Applied forecasting, NLP, and governed quantitative analysis to financial data, translating model evidence into decision support for investment and risk teams.

- Revamped XGBoost bond-market forecasting on rolling 24-month Bloomberg and internal data, using point-in-time features and horizon-specific evaluation to achieve a 62.4% directional hit rate.
- Built and deployed a FinBERT financial-news sentiment classifier with 87% reported accuracy and integrated the signal into credit-risk and investment-research workflows.
- Automated YODA and Bloomberg portfolio-data ingestion with schema, freshness, duplicate, reconciliation, logging, and idempotent-rerun controls, reducing manual effort by approximately 70%.
- Improved global mutual-fund forecasting accuracy by 33% through feature engineering and Bloomberg-data validation.
- Designed Python and SQL predictive pipelines to model equity performance and support quantitative research.

Accenture — Pune, India

Machine Learning Associate Jun 2018 – Jul 2021

Client: OP Financial, Finland

- Built credit- and debit-card limit decisioning and risk segmentation for OP Financial using account activity, loan history, credit scores, and policy constraints, contributing to 23% revenue growth and 40% operational-efficiency improvement.
- Designed risk-adjusted customer segments and A/B tests to measure campaign response, translating model outputs into targeted commercial actions.
- Applied t-tests, chi-square tests, ANOVA, and regression to identify portfolio patterns and quantify the business significance of observed outcomes.
- Built Tableau dashboards and SQL views tracking financial KPIs, improving market-data accuracy by 25%.

Client: Allianz Bank, Germany

- Developed an XGBoost churn classifier for Allianz (F1: 0.84), using precision, recall, threshold economics, and interpretable drivers to support retention decisions.
- Evaluated churn decisions with class-level metrics and operating-capacity trade-offs, keeping predictive feature importance separate from causal claims.
- Migrated enterprise data from DB2 and Teradata to AWS with validation and reconciliation controls.
- Standardized experimentation and validation practices to reduce bias and improve model interpretability.

PROJECTS

Financial Document Intelligence Agent

- Built a locally runnable agent over SEC filings with reranked retrieval, ticker-scoped evidence, and citations across five financial institutions.
- Built a Claude tool-calling router that separates filing search from read-only Text2SQL over XBRL facts and price history, with FastAPI, Docker, and Streamlit.
- Built a 15-question evaluation harness; SQL-backed numeric answers scored 5/5 for faithfulness, while narrative cases exposed a documented 2-4/5 limitation. Cloud deployment and full observability remain planned.

AI Quality Lab

- Built and published a React and TypeScript evaluation suite with six model-family workspaces, scenario controls, CSV evaluation, operating gates, monitoring, and downloadable evidence reports.
- Connected business value, model-quality metrics, operating constraints, and launch decisions in one browser-local product; the suite passes 33 automated tests.

Travel Marketplace Harm-Risk Prioritization

- Built a leakage-aware XGBoost system with chronological, label-mature splits and a locked 5% review policy; expected-loss ranking captured 37.19% of harmful value on an untouched future test versus 28.93% for probability ranking.

CERTIFICATIONS & CONTINUING EDUCATION

- Meta Data Analyst, issued by Meta
- Google Advanced Data Analytics, issued by Google
- Data Visualization in Tableau & Power BI, Udemy
- Continuing study in production LLM evaluation, observability, adversarial testing, and retrieval quality
- Q2 FY20 Accenture Team Award (Client & Customer category) — reduced technical issues by 84%

EDUCATION

Northeastern University Boston, MA

M.S. Data Analytics Engineering Sep 2021 – Dec 2023

Coursework: Data Management, Data Visualization, Statistical Methods, Data Mining, Machine Learning, Deep Learning

Narsee Monjee Institute of Management Studies Mumbai, India

B.Tech in Computer Engineering